

Answer key — teacher only

IGCSE Chemistry homework · 2.1–2.3 · CIE 0620 · 42 marks

Do not issue this file to students.**Section A — Multiple-choice (20 marks)**

Q	Ans	Source	Q	Ans	Source
A1	B	2023 MJ P23 Q2	A11	A	2024 MJ P21 Q5
A2	C	2023 MJ P21 Q2	A12	D	2023 MJ P22 Q2
A3	A	2025 ON P22 Q4	A13	B	2021 MJ P22 Q8
A4	B	2024 MJ P21 Q4	A14	B	2023 MJ P21 Q3
A5	B	2024 ON P21 Q3	A15	C	2024 MJ P22 Q3
A6	A	2023 ON P23 Q3	A16	B	2025 ON P21 Q5
A7	C	2025 ON P21 Q3	A17	C	2025 ON P23 Q4
A8	B	2021 ON P22 Q4	A18	B	2023 ON P22 Q4
A9	B	2025 MJ P22 Q4	A19	B	2025 MJ P21 Q4
A10	B	2023 MJ P21 Q5	A20	C	2025 ON P23 Q6

Quick string: B C A · B B A C B · B B A D · B B C B C · B B C

Section B — Mark scheme clips**B1 · 2021 ON Paper 42 Question 3 (8 marks)****B1(a) [3]**

Question	Answer	Marks												
3(a)	1 mark for each correct row <table> <tr> <th>Name</th><th>Relative mass</th><th>Relative charge</th></tr> <tr> <td>Proton</td><td>1</td><td>+1</td></tr> <tr> <td>Neutron</td><td>1</td><td>0</td></tr> <tr> <td>Electron</td><td>1 / 1840</td><td>−1</td></tr> </table>	Name	Relative mass	Relative charge	Proton	1	+1	Neutron	1	0	Electron	1 / 1840	−1	3
Name	Relative mass	Relative charge												
Proton	1	+1												
Neutron	1	0												
Electron	1 / 1840	−1												

B1(b) [5]

3(b)

Particle	Number of protons	Number of neutrons	Number of electrons
$^{32}_{16}\text{S}$	16	16	16
$^{39}_{19}\text{K}^+$	19	20	18
$^{79}_{35}\text{Br}^-$	35	44	36

M1 = row 1 (1) M2 = row 2 (1) M3 = Br (1)
M4 = $^{79}_{35}$ (on left of any symbol) (1) M5 = charge (on any symbol) (1)

5

B2 · 2024 ON Paper 41 Question 2 (14 marks)

B2(a) [1]

2(a)	total number of protons and neutrons in the nucleus of an atom	1
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B2(b) [1]

2(b)	they are the same	1
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B2(c) [2]

2(c)	M1 2,8,4 M2 2,8,8	2
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B2(d) [5]

2(d)	M1 $12n + 11e$ M2 10e M3 Ga M4 69 above 31 and to the left of the symbol M5 charge of 3 +	5
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B2(e)(i) [2]

2(e)(i)	M1 atom(s) of the same element M2 with different number of neutrons	2
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B2(e)(ii) [2]

Question	Answer	Marks
2(e)(ii)	M1 (3×203) AND (7×205) (= 2044) M2 204.4	2

B2(e)(iii) [1]

2(e)(iii)	same number of electrons / same electronic configuration	1
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Question	Answer	Marks
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